Polynomial Regression

A Polynomial regression is used when the relationship between X and Y is curved, not a straight line.

→ linear regression draws a straight line

→ Polynomial regression draw a Curve

But internally, it is a linear regression

why do we need Polynomial Regression:-

Linear regression Assumes: When X increases by 1 unit, Y increases (decreases) by a constant amount.

→ But in real life, this is oftne not True

Example:- • Salary vs experience: Salaray increases fast initially, then slows down.

• Sales vs advertising spend:- more ads helps, but after a Point returns reduce

→ These relationships are curved, not straight, So linear regression fails

→ Polynomial regression does not change the algorithm

→ It only creates new features

Original feature: X

Polynomial feature: X, X^2, X^3

→ Linear regression : $\boxed{Y = \beta_0 + \beta_1 X}$

→ Polynomial regression (degree = 2) : $\boxed{Y = \beta_0 + \beta_1 X + \beta_2 X}$

→ Polynomial regression (degree = 3) : $\boxed{Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3}$

- The curve comes because of X^2, X^3 not because of

Why it is still called a Linear regression :-

Because : → β_0, β_1, β_2 appear linearly

→ No β is squared or multiplied with another β

So : → linear in Parameters

→ Solved using same OLS formula

Choosing the degree

- degree 1 → Linear regression
- degree 2 → Simple curve
- degree 3+ → more bends